AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 7 · STUDENT LESSON

Hypotheses and measurable outcomes

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

Before you start

Recall A literature comparison matrix. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of hypotheses and measurable outcomes. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of hypotheses and measurable outcomes.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from A literature comparison matrix. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Methods results and limitations. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from A question that **evidence** can answer. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of hypotheses and measurable outcomes

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall A literature comparison matrix. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain hypotheses and measurable outcomes using the records and boundaries of this fictional investigation.

Notice the evidence

> Investigate the supplied Hypotheses and measurable outcomes case. Separate observations, given facts and assumptions before choosing an explanation.

A **hypothesis** predicts a measurable comparison under stated conditions. Declare **primary outcome** and direction before inspecting results.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Hypotheses and measurable outcomes. Connect each labelled input, deciding step and result.

Toy **hypothesis**:method B improves supported-answer rate over baseline A on a reserved paired set. Define support rubric and failure treatment; record secondary outcomes separately.

Hypotheses and measurable outcomes — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
After seeing faster responses, the team changes its primary claim from support improvement to speed improvement.	Toy hypothesis:method B improves supported-answer rate over baseline A on a reserved paired set. Define support rubric and failure treatment; record secondary outcomes separately.	Report the change as exploratory/post-result revision. Preserve the original support hypothesis; do not present speed as the prespecified primary outcome.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Hypotheses and measurable outcomes. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

After seeing faster responses, the team changes its primary claim from support improvement to speed improvement. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: B sounds friendlier but has the same unsupported count as A.

Predicted result

Prediction: The stated improvement **hypothesis** is not supported on these cases.

Why the result follows

Reason: Friendliness is a different outcome from the predefined unsupported-case count.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Hypotheses and measurable outcomes to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Hypothesis	A stated expectation to test
Primary outcome	The prespecified main measure
Exploratory result	An observation or analysis chosen after seeing evidence

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Hypotheses and measurable outcomes, predict a step, then compare with its explanation.

After seeing faster responses, the team changes its primary claim from support improvement to speed improvement.

Report the change as exploratory/post-result revision. Preserve the original support **hypothesis**; do not present speed as the prespecified **primary outcome**.

Worked explanation · Hypotheses and measurable outcomes

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

> The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Hypotheses and measurable outcomes

Write a testable **hypothesis** before scoring. A team predicts that a retrieval rule will reduce unsupported answers on a fixed validation set.

The facts we will use

> #### Supplied facts
>
> Ten cases are reserved. Unsupported means a factual claim absent from the approved card. **Prediction**: B has fewer unsupported cases than A.

AI ACADEMY / AI-7/7

Hypotheses and measurable outcomes

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name the directional claim

The prediction concerns unsupported cases, not writing elegance.



2. Define scoring first

Each answer either contains an unsupported claim or does not under the fixed rubric.



3. Freeze the comparison

Run both versions on the same cases after recording the hypothesis.

CHANGE ONE THING

B sounds friendlier but has the same unsupported count as A.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Name the directional claim	The prediction concerns unsupported cases, not writing elegance.
2. Define scoring first	Each answer either contains an unsupported claim or does not under the fixed rubric.
3. Freeze the comparison	Run both versions on the same cases after recording the hypothesis.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **hypothesis** makes an expected pattern explicit before seeing outcomes. The measured definition must be clear enough for another reviewer to apply.

What this example does not establish

Keep the stated limit

No power calculation or significance claim is provided; a null descriptive difference is not proof of exact equality.

Keep the stated limit

Read the labelled picture: Name the predicted outcome and how unsupported is scored. If B sounds friendlier but ties A's unsupported count, does the recorded improvement **hypothesis** receive support?

Keep the stated limit

Change one thing: B sounds friendlier but has the same unsupported count as A. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Hypotheses and measurable outcomes claim and explain a repair using the supplied evidence.

A **hypothesis** can fail honestly; changing it after results does not erase the original test.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: A **hypothesis** can fail honestly; changing it after results does not erase the original test. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Hypotheses and measurable outcomes with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Hypotheses and measurable outcomes”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Hypotheses and measurable outcomes. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Name the predicted outcome and how unsupported is scored. If B sounds friendlier but ties A's unsupported count, does the recorded improvement **hypothesis** receive support?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Hypotheses and measurable outcomes. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

After seeing faster responses, the team changes its primary claim from support improvement to speed improvement. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or "not run"	

Record	Your evidence
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Hypotheses and measurable outcomes. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: A **hypothesis** can fail honestly; changing it after results does not erase the original test. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Reproducible AI Research Study. Keep an annotated decision and the evidence supporting it.

Add a labelled hypotheses and measurable outcomes **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of hypotheses and measurable outcomes.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Hypotheses and measurable outcomes case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Hypotheses and measurable outcomes — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
After seeing faster responses, the team changes its primary claim from support improvement to speed improvement.	Toy hypothesis:method B improves supported-answer rate over baseline A on a reserved paired set. Define support rubric and failure treatment; record secondary outcomes separately.	Report the change as exploratory/post-result revision. Preserve the original support hypothesis; do not present speed as the prespecified primary outcome.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Before running12 cases, you expect B support to exceed A. Write primary metric and one falsifying result.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Before running12 cases, you expect B support to exceed A. Write primary metric and one falsifying result.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Hypotheses and measurable outcomes, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Write a testable **hypothesis** before scoring. A team predicts that a retrieval rule will reduce unsupported answers on a fixed validation set.

By the end, I can	How I will show it
Explain the mechanism and terminology of hypotheses and measurable outcomes	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Hypotheses and measurable outcomes without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of hypotheses and measurable outcomes

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building

> block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Write a testable **hypothesis** before scoring. A team predicts that a retrieval rule will reduce unsupported answers on a fixed validation set.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Run both versions on the same cases after recording the **hypothesis**.

1. Name the directional claim

2. Freeze the comparison

3. Define scoring first

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Freeze the comparison Run both versions on the same cases after recording the **hypothesis**. A **hypothesis** makes an expected pattern explicit before seeing outcomes.

The measured definition must be clear enough for another reviewer to apply.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: B sounds friendlier but has the same unsupported count as A. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. The stated improvement **hypothesis** is not supported on these cases.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The stated improvement **hypothesis** is not supported on these cases. Friendliness is a different outcome from the predefined unsupported-case count.

Repair a missing building block

Review A literature comparison matrix (ai-7/6). Goal: Explain the **mechanism** and terminology of a literature comparison matrix.

Short practice: Explain why 8/10 indoors and 16/20 outdoors share a rate but not a controlled comparison. Which condition column must the literature matrix preserve?. Return to this week and explain what you changed.

Review Methods results and limitations (ai-7/4). Goal: Explain the **mechanism** and terminology of methods results and limitations.

Short practice: Place the five-photo procedure, 3/5 versus 4/5 finding and missing outdoor **evidence** in their proper categories. Why cannot the result establish that B is best for outdoor farming?.

Return to this week and explain what you changed.

Review A question that **evidence** can answer (ai-7/1). Goal: Explain the **mechanism** and terminology of a question that **evidence** can answer. Short practice: Which proposed diagram question has a measurable outcome?

Name its versions, six-case denominator and fixed scoring, then explain why giving only B easier tasks spoils the comparison.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

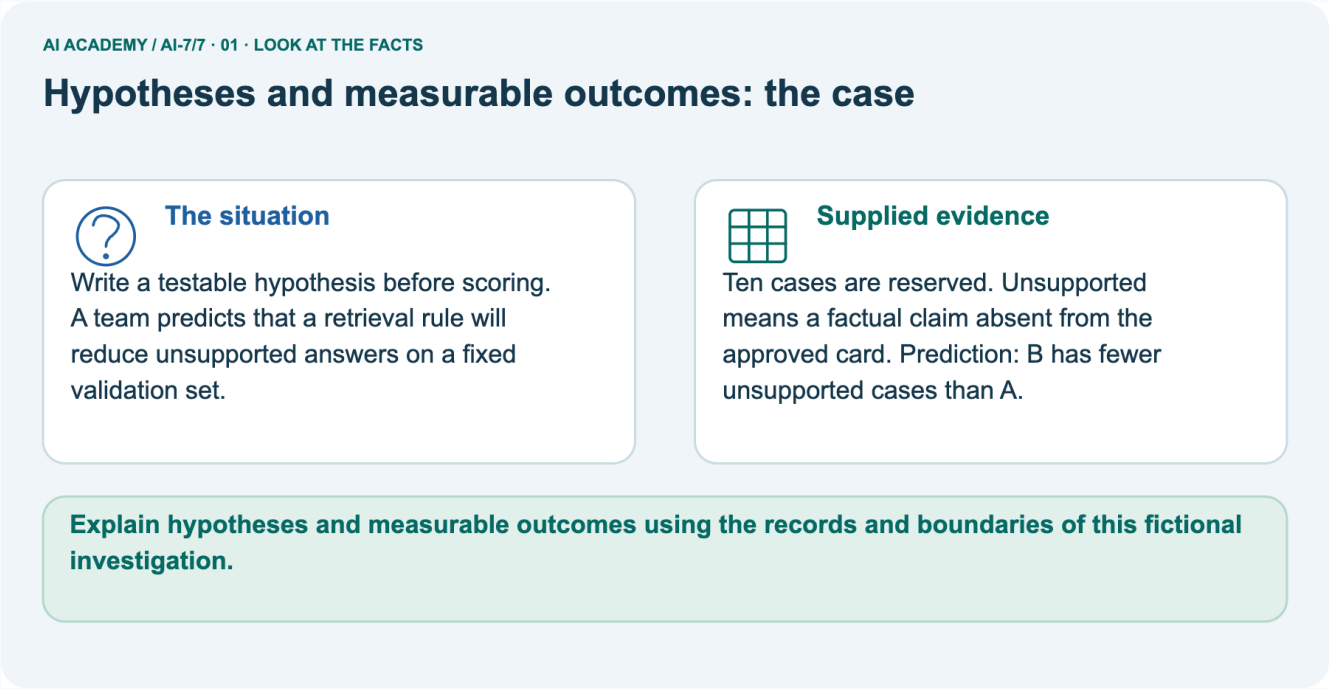
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Hypotheses and measurable outcomes graphic using its labels, deciding evidence and relevant limitation.

01. Hypotheses and measurable outcomes: the case



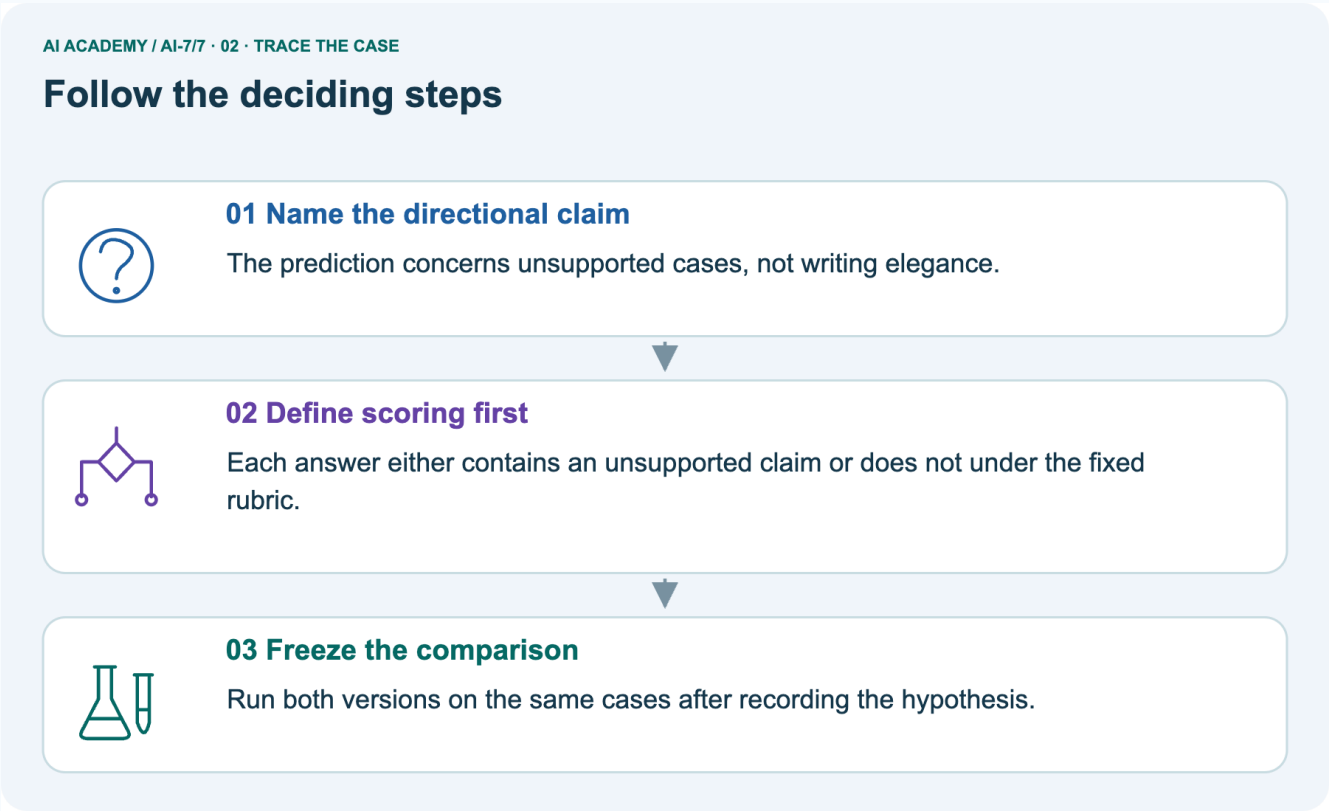
Follow the labelled picture with the matching explanation. Open the vector diagram

Write a testable **hypothesis** before scoring. A team predicts that a retrieval rule will reduce unsupported answers on a fixed validation set. Supplied facts: Ten cases are reserved.

Unsupported means a factual claim absent from the approved card. **Prediction**: B has fewer unsupported cases than A.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

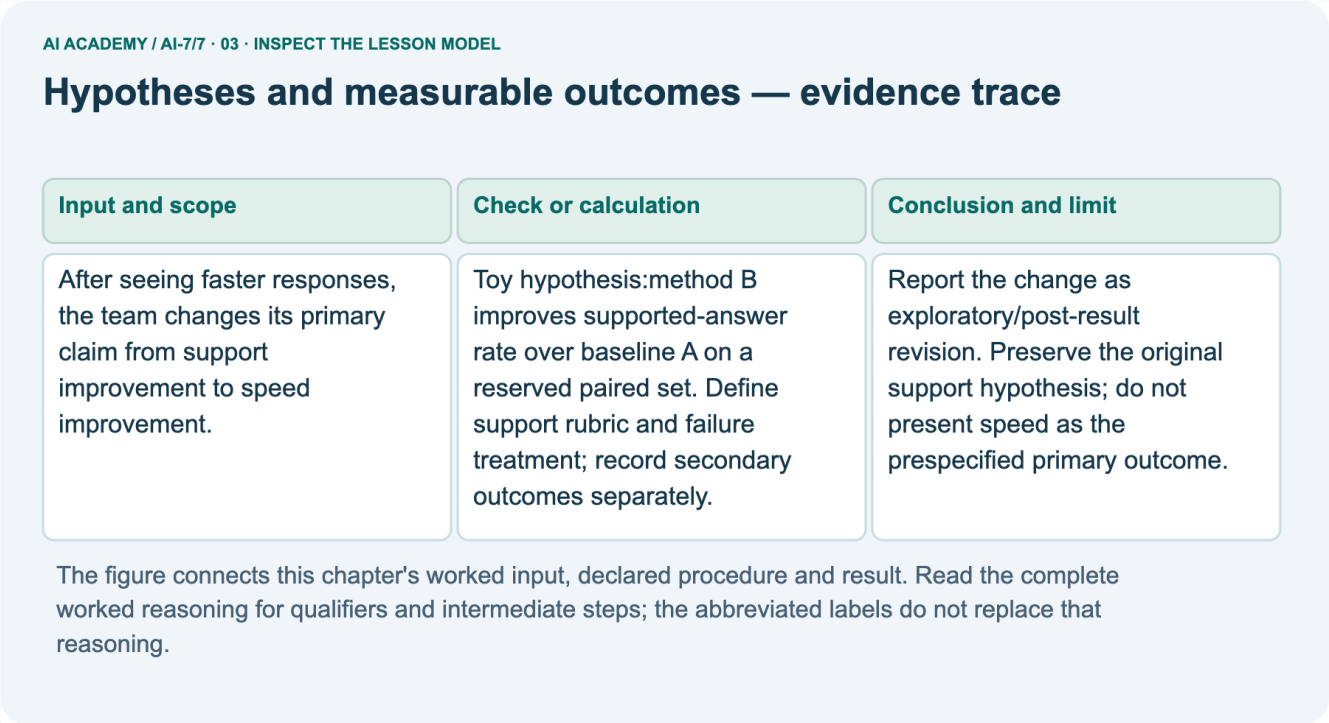


Follow the labelled picture with the matching explanation. Open the vector diagram

Name the directional claim: The **prediction** concerns unsupported cases, not writing elegance. → Define scoring first: Each answer either contains an unsupported claim or does not under the fixed rubric. → Freeze the comparison: Run both versions on the same cases after recording the **hypothesis**.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Hypotheses and measurable outcomes — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Hypotheses and measurable outcomes — **evidence** trace.

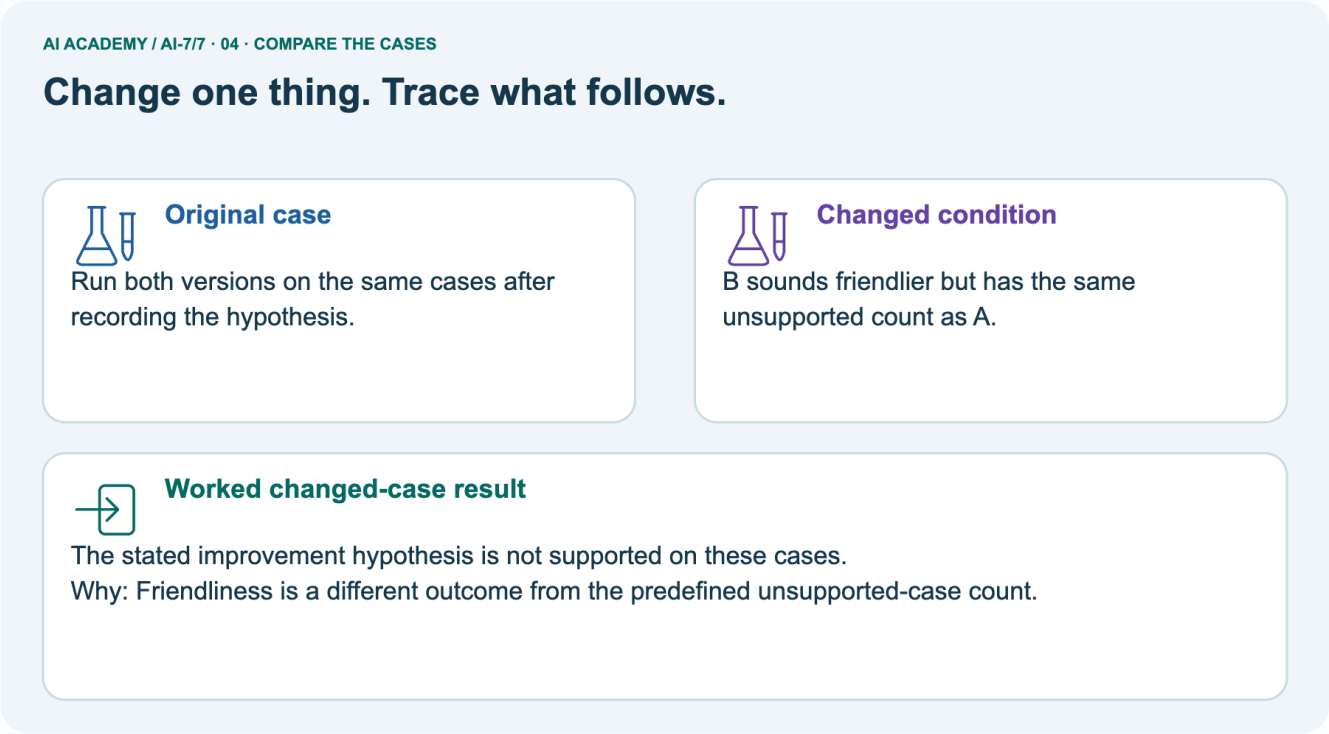
After seeing faster responses, the team changes its primary claim from support improvement to speed improvement. | Toy **hypothesis**:method B improves supported-answer rate over baseline A on a reserved paired set.

Define support rubric and failure treatment; record secondary outcomes separately. | Report the change as exploratory/post-result revision.

Preserve the original support **hypothesis**; do not present speed as the prespecified **primary outcome**.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



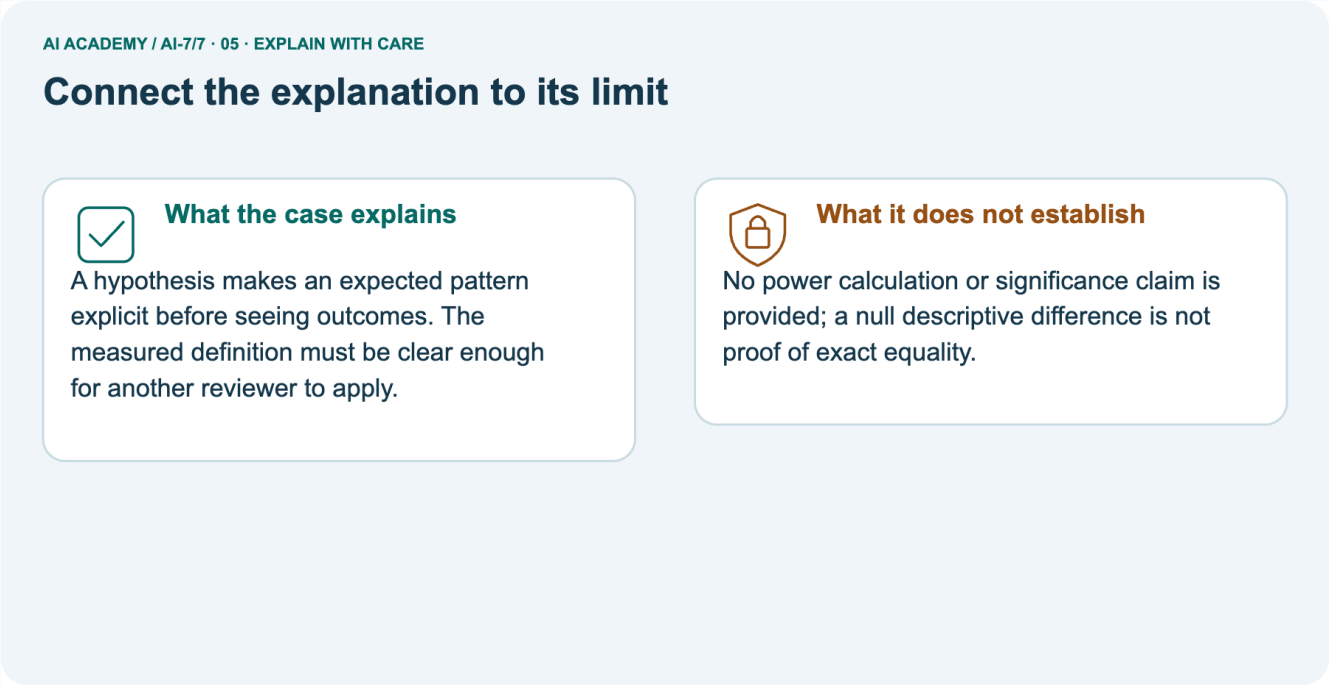
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Run both versions on the same cases after recording the **hypothesis**. Changed condition: B sounds friendlier but has the same unsupported count as A.

Expected result: The stated improvement **hypothesis** is not supported on these cases. Reason: Friendliness is a different outcome from the predefined unsupported-case count.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



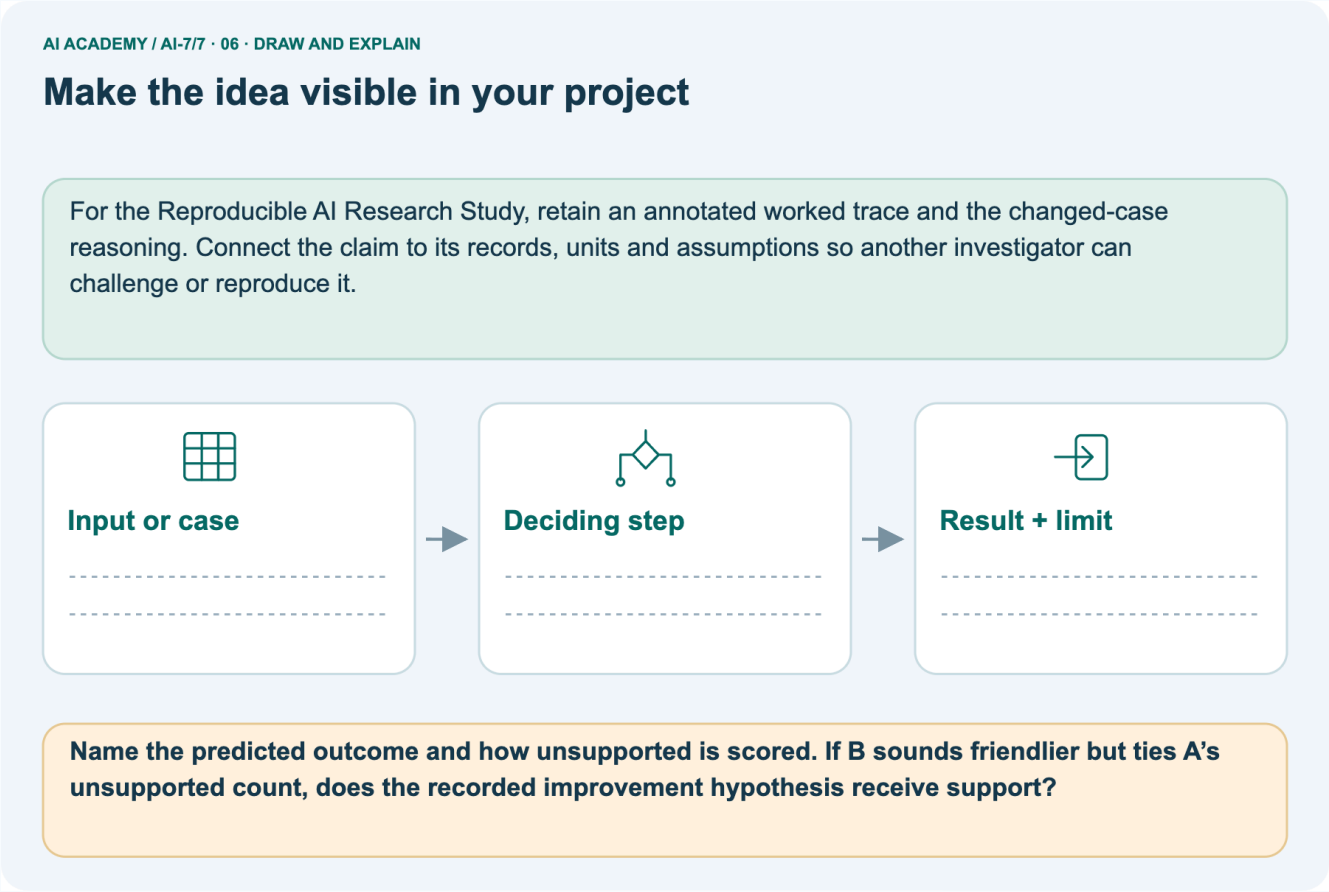
Follow the labelled picture with the matching explanation. Open the vector diagram

A **hypothesis** makes an expected pattern explicit before seeing outcomes. The measured definition must be clear enough for another reviewer to apply.

Limit: No power calculation or significance claim is provided; a null descriptive difference is not proof of exact equality.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Name the predicted outcome and how unsupported is scored.

If B sounds friendlier but ties A's unsupported count, does the recorded improvement **hypothesis** receive support?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.